

10(a)	<i>any eight from:</i> 1 fibres are multinucleate ; 2 cell surface membrane is sarcolemma ; 3 sarcoplasm has many mitochondria ; 4 sarcoplasmic reticulum membranes have protein pumps ; 5 transverse system tubules / T-system ; 6 <i>ref. to</i> myofibrils ; 7 thick filament / myosin, attached to M line ; 8 thin filament / actin, attached to Z line ; 9 interdigitation of filaments causes striated appearance ; 10 description of one of, A / H / I, bands ; 11 sarcomere is the distance between M lines ; 12 myosin is a fibrous protein with globular protein head ; 13 actin is a chain of globular protein molecules ; 14 tropomyosin / troponin, attached to actin ; <i>allow marks from an annotated diagram</i>	8
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Question	Answer	Marks
10(b)	<i>any seven from:</i> 1 acid-growth (hypothesis) ; 2 auxin stimulates proton pumps ; 3 (in) cell surface membrane ; 4 protons / H⁺, pumped into cell wall ; 5 using energy / by active transport ; 6 pH of cell wall decreases / cell wall becomes (more) acidic ; 7 pH-dependent enzymes activated ; 8 <i>ref. to</i> expansins ; 9 bonds between cellulose microfibrils broken ; 10 <i>idea that</i> cell wall, 'loosens' / becomes more elastic / able to stretch ; 11 (more) water enters cell / turgor pressure increases ; 12 (so) cell (wall) expands ;	7